Statistical Methods in Education Research
Vanderbilt University
Prof. Sean P. Corcoran

November 3, 2026
63 total points

Problem Set 8 Solutions

---

1. Data are gathered on 16 students. Half of the students are randomly assigned to a new tutoring program and half have their usual schooling experience. A study finds that test scores for the tutoring program students are 10 points higher on average than those for the other students (p=*0.3*). The mean test score overall is 200 with a standard deviation of 40. (**3 points—1 each**)

   (a) Is the study's finding statistically significant? Explain why or why not.

   > No, the p-value from a test of differences in means is reported to be 0.3, above most thresholds for statistical significance.

   (b) Is the study's finding practically significant, in your opinion? Explain what practical significance means here.

   > To assess practical significance, we can compare the 10 point average gain from tutoring to the overall standard deviation on the exam of 40, for Cohen's $d = 10/40 = 0.25$. According to the review by Kraft, a 0.25 standard deviation difference is on the large side for educational interventions.

   (c) Would you conclude that the tutoring program is effective based on these results? Ineffective? Briefly explain.

   > No, the results are inconclusive. We cannot rule out large beneficial effects of tutoring, but the estimated impact is statistically insignificant and may have occurred by chance. A larger sample would help us determine whether the effect is real or not.

2. Two studies were commissioned to evaluate an intensive program designed to enhance social and emotional learning (SEL) among adolescents. The index used to measure SEL has a mean of 50 and a standard deviation of 10. Study 1 failed to find a statistically significant improvement in SEL, with a 95% confidence interval for the gain in SEL of $(-7, 17)$. Study 2 also failed to find a statistically significant improvement in SEL, with a 95% confidence interval of $(-2.5, 1.5)$. Which of these two studies (if any) is more informative to a policymaker, in your opinion, and why? (**4 points**)

In both cases the confidence interval contains zero, so neither provides evidence of a statistically significant impact of the program on SEL. However, the 95% confidence interval in Study 2 is much narrower: $(-2.5, 1.5)$. If the confidence interval provides a range of null hypotheses that our data are consistent with (cannot reject), then Study 2 can largely rule out large positive (or negative) effects of the program on SEL. (At the upper bound, a +1.5 effect would be a $1.5/10 = 0.15$ standard deviations, a modestly large effect). The confidence interval in Study 1 ranges from -7 to 17, encompassing both large negative and large positive effects; it provides little guidance to a policymaker.

3. A decision is planned in a test of $H_0 : \mu = 0$ against the alternative $H_a : \mu > 0$, using $\alpha = 0.05$. If the actual μ is 5, the probability of a Type II error (β) is 0.17. (**6 points—2 each**)

   (a) Explain the meaning of the last sentence in words. (**2 points**)

   The probability of a Type II error (β) is the probability one does not reject H_0 when it is false. Put another way, in 17% of random samples this test will not reject H_0 if μ is 5 rather than 0.

   (b) If the test used $\alpha = 0.01$, would the probability of a Type II error be less than, equal to, or greater than 0.17? Explain. (**2 points**)

   With an $\alpha = 0.01$ rather than $\alpha = 0.05$, you reject H_0 less often. Since β is the probability one does not reject H_0 when it is false, a lower α <u>increases</u> β.

   (c) If μ were actually 10, would the probability of a Type II error be less than, equal to, or greater than 0.17? Explain. (**2 points**)

   If the true $\mu = 10$, it is much less likely that the test will fail to reject $H_0 : \mu = 0$. This means a <u>lower</u> Type II error rate.

4. Your team is tasked with studying whether charter school students perform *better* in mathematics, on average, than traditional public school students. You know that math scores in the population of public school students have a normal distribution with a mean of 420 and standard deviation of 84. You plan to administer the same test to a sample of charter school students to test your hypothesis. Use Stata or a web applet to answer the following questions. (**14 points**)

   (a) What are the null and alternative hypotheses in this problem? (**2 points**)

Let μ be the mean test performance of charter school students in the population.

$$H_0 : \mu = 420$$

$$H_1 : \mu > 420$$

(b) Describe what a Type I and Type II error would be in this study. (**2 points**)

A Type I error is rejecting H_0 when it is true. In this example, it would mean concluding that charter school students perform better, on average, than traditional public school students, when in fact they do not. A Type II error is *not* rejecting H_0 when it is false. In this example, it would mean concluding that charter school students perform no better, on average, than traditional public school students, when in fact they do.

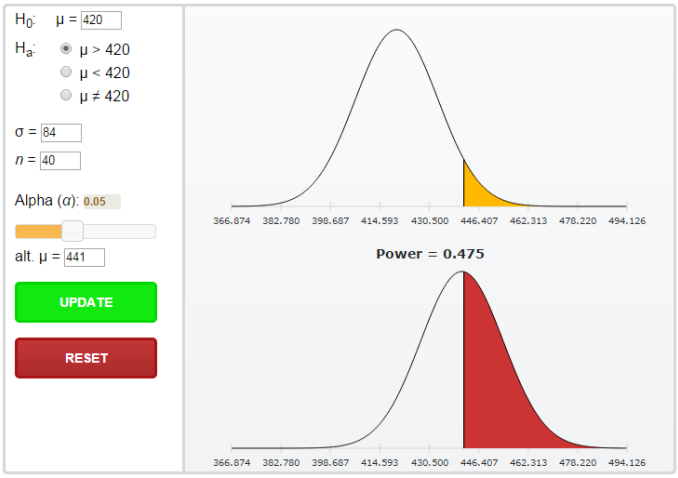
(c) You consider a meaningful difference in test scores to be +21 points. What specific alternative μ_a would this imply? How large an *effect size* (Cohen's d) would this be? (**4 points**)

A 21 point difference in test scores would mean an alternative μ of $420 + 21 = \mathbf{441}$. In terms of an effect size (using Cohen's d), this would be: $(21/84) = \mathbf{0.25}$ sd units. In other words, charter school students would be performing 0.25 standard deviations better than non-charter students if their mean score was 441.

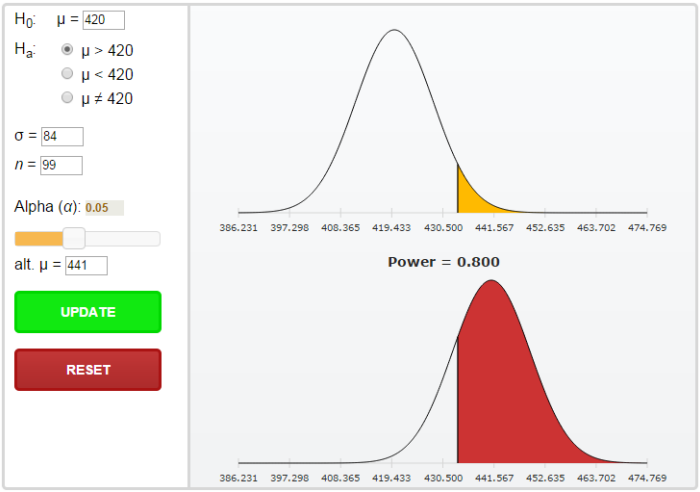
(d) If you were to randomly sample 40 charter school students to take the math test, what is the power associated with your significance test (assuming the alternative in part c)? What is the probability of a Type II error? Use significance level $\alpha = 0.05$. (**3 points**)

Using the applet, the power of the test $(1 - \beta)$ would be **0.475**. The probability of a Type II error (β) would be **0.525**. In words, there is a 0.525 probability of not rejecting H_0 when μ is actually 441. This is a fairly high probability of not detecting a rather large effect size. If the true difference between charter and non-charter students were 21 points (0.25 sd), one would hope that a study design would be able to identify such effects. To put the power of the test in words, there is a 0.475 probability of rejecting H_0 when μ is actually 441 (again, not a very high probability).

(e) Holding everything else constant, how large of a sample would you need in order to increase the power to 0.80? (**3 points**)



One way to determine this is through trial and error with the power applet. Keep increasing n until the power rises to 0.80. The required sample size is **99** or more students (see below). You could also use Stata's `power` command to find the minimum sample size (also shown below).




```
. power onemean 420 441, sd(84) knownsd onesided

Estimated sample size for a one-sample mean test
z test
Ho: m = m0  versus  Ha: m > m0

Study parameters:

        alpha =    0.0500
        power =    0.8000
```

```
        delta =    0.2500
           m0 =  420.0000
           ma =  441.0000
           sd =   84.0000

Estimated sample size:

            N =        99
```

5. Your favorite candidate currently has a favorability rating of 49%. You plan to take a random sample of 100 likely voters to assess whether her favorability has *improved* (i.e., is greater than 49%). Use Stata to answer the following questions, and use $\alpha = 0.05$. (Hint: you can use the power calculator for proportions called `power oneproportion`): (**8 points**)

   (a) Write down the null and alternative hypotheses for this test. (**2 points**)

   $H_0 : \pi = 0.49$ and $H_1 : \pi > 0.49$

   (b) What is the statistical power of your test if the true favorability rating is 55%? Provide an explanation in words. (**2 points**)

   See Stata output below—the power is only 0.3275. In words, over repeated samples, H_0 will only be (correctly) rejected 33% of the time when the true favorability is 0.55.

```
. power oneproportion 0.49 0.55, n(100) onesided

Estimated power for a one-sample proportion test
Score z test
Ho: p = p0  versus  Ha: p > p0

Study parameters:

        alpha =    0.0500
            N =       100
        delta =    0.0600
           p0 =    0.4900
           pa =    0.5500

Estimated power:

        power =    0.3275
```

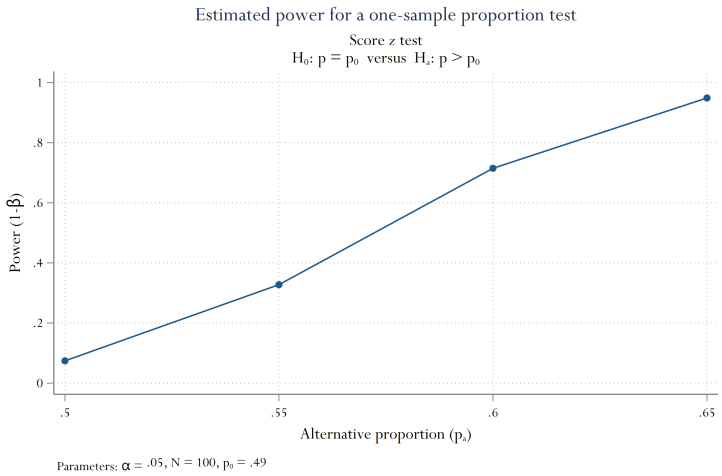
   (c) Have Stata produce a table and graph of power for alternative hypotheses ranging from 50% to 65%, in increments of 5%. What is the smallest alternative that can be detected with 80% power? (**4 points**)

See Stata output below. Use the figure, the smallest alternative that will be rejected with 80% power is about 62-63% favorability.

```
. power oneproportion 0.49 (0.50(0.05)0.65), n(100) onesided

Estimated power for a one-sample proportion test
Score z test
Ho: p = p0  versus  Ha: p > p0
```

alpha	power	N	delta	p0	pa
.05	.0743	100	.01	.49	.5
.05	.3275	100	.06	.49	.55
.05	.7146	100	.11	.49	.6
.05	.9485	100	.16	.49	.65



6. A recent paper by von Hippel (2025) argues that effect sizes expressed in standard deviation units are not easily understood by educators (see also Lortie-Forgues, Sio, and Inglis 2021 on this point). He recommends that these be converted to more relatable units, such as percentile points. A challenge with this is the percentile point change implied by a delta sd effect depends on the starting point. von Hippel shows that–for a wide range of values–you can get a pretty good approximation of the percentile point change by multiplying the effect size in sd units by 37. Explore these ideas using simulated data in Stata following the steps below. (**10 points**)

   - Create an empty dataset with 1000 observations.

- Create a new variable $x1$ that contains random draws from the standard normal distribution. We will assume the outcome of interest (e.g., a test score) follows this distribution.

- Create a new variable $x1p$ that contains the percentile of the standard normal distribution that each value of $x1$ represents (Hint: how can we find the percent of all values in a standard normal that fall below a particular value $x1$)?

- Suppose an educational intervention has a 0.25 sd effect on all student scores. Create a new variable $x2$ that equals $x1 + 0.25$. According to the von Hippel rule of thumb, what is the approximate effect in percentage point terms?

- Create a new variable $x2p$ that contains the percentile rank of the standard normal distribution that each value of $x2$ represents and then a new variable *chgp* that contans the change in percentile rank ($x2p - x1p$).

- Create a scatter plot (`scatter y x`) that plots the *chgp* against the starting score $x1p$. What do you observe?

- Consider the starting points $x1p$ in the interval [0.2, 0.8]—i.e. the middle 3/5 of the distribution. Summarize the *chgp* in this interval. How closely does it correspond to the von Hippel rule of thumb? (You may also see this in the scatter).

See attached Stata output. von Hippel's approximate percentile point change for a 0.25 sd effect is $37 \times 0.25 = 9.25$. Looking at the scatterplot, the percentile point change ranges between 8 and 10 for a large fraction of values (about half or more). The percentile point change gets further away (and smaller) from 9-10 the lower or higher the starting pecentile rank. So, the approximation works fairly well for starting values in the middle of the distribution. (This can also be seen in the descriptive statistics. For the full range of data, the mean/median percentile point change is 7.05/8.00. For the middle 3/5 of the distribution, the mean/median change is 8.92/9.23, a lot closer to 9.25).

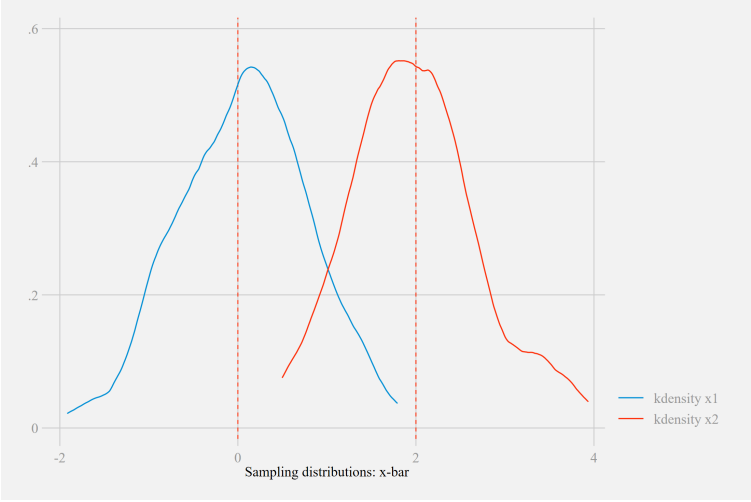
7. The Stata syntax below will create a dataset with 200 sample means for $\bar{x}_1$ and $\bar{x}_2$, and 200 sample standard deviations s_1 and s_2. Each sample consists of $n = 50$ random draws from normal distributions in which $x_1 \sim N(0, 5)$ and $x_2 \sim N(2, 5)$. (That is, x_1 has a normal distribution with mean zero and standard deviation 5; x_2 has a normal distribution with mean 2 and standard deviation 5). (**18 points**)

   (a) Run the Stata syntax above, and choose your own starting seed value (#). Take care that your quotations are correct, as they may not copy over correctly from

this document into a do-file. Use the `twoway` graphing function with `kdensity` to overlay the resulting distributions of $\bar{x}_1$ and $\bar{x}_2$ (see below). `kdensity` is a kernel density function, a kind of "smoothed" histogram. (**3 points**)

```
set seed 1989

forvalues j=1/200 {
   clear
   tempfile sample‘j’
   set obs 50
   gen x1 = rnormal(0,5)
   gen x2 = rnormal(2,5)
   collapse (mean) x1 x2 (sd) sx1=x1 sx2=x2
   gen n= 50
   save ‘sample‘j’’
   }
use ‘sample1’, clear
forvalues j=2/200 {
   append using ‘sample‘j’’
   }
twoway (kdensity x1) (kdensity x2), xtitle(Sampling distributions: x-bar) ///
   xline(0) xline(2)
```



(b) Create two new variables that contain the lower and upper bounds of a 95% confidence interval for μ based on $\bar{x}_1$. (Each resulting $\bar{x}_1$—all 200 of them—has a confidence interval associated with it). Be sure to use the sample standard deviation when creating your confidence intervals, not the population standard deviation, and continue to do so in part (c). In what percentage of samples does your confidence interval contain the true population mean $\mu = 0$? In what percentage does it not? (Hint: you can create a third variable that flags whether the CI contains zero or doesn't). (**3 points**)

Stata syntax and results below. In my case, 93% of random samples result in confidence intervals that contain the μ of 0 (and in 7% they do not). Notice the confidence interval is constructed using the t-distribution, since I am using s rather than the known standard deviation σ.

```
// confidence intervals
gen lb1 = x1 - invttail(49, 0.025)*(sx1/sqrt(50))
gen ub1 = x1 + invttail(49, 0.025)*(sx1/sqrt(50))

// contains mu=0
gen contains01 = (lb1<0 & ub1>0)
tabulate contains01
```

contains01	Freq.	Percent	Cum.
0	14	7.00	7.00
1	186	93.00	100.00
Total	200	100.00	

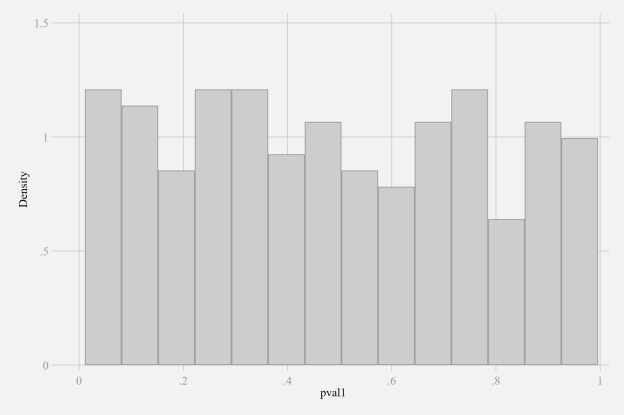
(c) Create two new variables that equal the t-statistics and p-values from a test of $H_0 : \mu = 0$ against $H_1 : \mu \neq 0$. (Each resulting $\bar{x}_1$ will have a t and p-value for this test). Note this is a two-sided test. Create a histogram to show the resulting distribution of p-values across the 200 random samples. In what percentage of your random samples is $p < 0.05$? Explain in words what this percentage represents. (**4 points**)

Stata syntax and results below. In my case, 7% of random samples have a $p < 0.05$, and 93% do not. In a two-tailed test this should produce the same conclusion as the 95% confidence interval test in part (b). Notice the p-value is found with reference to the t-distribution, since I am using s rather than the known standard deviation σ. These 7% of random samples are instances in which one improperly rejects H_0—a Type I error.

```
// tstat and pvalue
gen tstat1 = (x1-0)/(sx1/sqrt(50))
gen pval1 = 2*ttail(49, abs(tstat1))
gen sig1 = pval1<=0.05
tabulate sig1
histogram pval1
```

sig1	Freq.	Percent	Cum.
0	186	93.00	93.00
1	14	7.00	100.00

```
------------+-----------------------------------
      Total |        200      100.00
```



(d) You are still interested in the null hypothesis $H_0 : \mu = 0$. However, suppose that—unbeknownst to you—the sample means $\bar{x}_2$ (not the $\bar{x}_1$ that you used in parts b-c) reflect draws from the true distribution of x. That is, x has a mean of 2, not 0. As in part (b), create two new variables that contain the lower and upper bounds of a 95% confidence interval for μ, based on $\bar{x}_2$ (and s_2). In what percentage of random samples does your confidence interval based on $\bar{x}_2$ contain *zero*? Explain in words what this percentage represents. (**4 points**)

Stata syntax and results below. In my case, 23% of random samples result in confidence intervals that contain the hypothesized mean of 0 (and in 77% they do not). This 23% represents a Type II error: failing to reject $H_0 : \mu = 0$ when it is false.

```
// now using x2

// confidence intervals
gen lb2 = x2 - invttail(49, 0.025)*(sx2/sqrt(50))
gen ub2 = x2 + invttail(49, 0.025)*(sx2/sqrt(50))

// contains zero
gen contains02 = (lb2<0 & ub2>0)
tabulate contains02

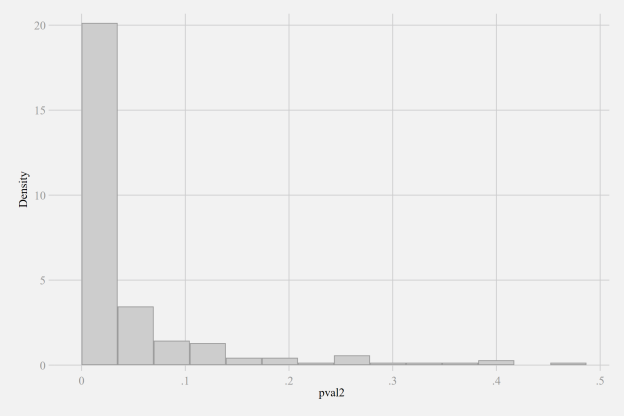
 contains02 |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |        154       77.00       77.00
          1 |         46       23.00      100.00
------------+-----------------------------------
      Total |        200      100.00
```

(e) Repeat part (c) using $\bar{x}_2$ and continue to test the hypothesis $H_0 : \mu = 0$ against $H_1 : \mu \neq 0$. In what percentage of these random samples is $p < 0.05$? In what percentage of these random samples is $p > 0.05$? Explain in words what these percentages represent. (**4 points**)

Stata syntax and results below. In my case, 77% of random samples have a $p < 0.05$, and 23% do not. Again, this 23% represents a Type II error: failing to reject $H_0 : \mu = 0$ when it is false.

```
// tstat and pvalue
gen tstat2 = (x2-0)/(sx2/sqrt(50))
gen pval2 = 2*ttail(49, abs(tstat2))
gen sig2 = pval2<=0.05
tabulate sig2
histogram pval2
```

```
       sig2 |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |         46       23.00       23.00
          1 |        154       77.00      100.00
------------+-----------------------------------
      Total |        200      100.00
```



```
.
. // ************************************************************************
. // LPO.8800 Problem Set 8 - Solutions
. // Last updated: October 30, 2026
. // ************************************************************************
.
. // *************
. // Question 6
. // *************
.
. clear

. set seed 1973

.
. // create dataset with 1000 observations
. set obs 1000
Number of observations (_N) was 0, now 1,000.

. // random draws from standard normal (x1)
. gen x1= rnormal()

. // percentile in the standard normal (x1p)
. gen x1p=normal(x1)*100

.
. // apply a 0.25 "effect" to every x1 score values
. gen x2= x1 + 0.25

. // von Hippel approximation of change in percentile rank (x37)
. di 37*0.25
9.25

. // percentile in the standard normal (x2p)
. gen x2p=normal(x2)*100

. // change in percentile rank from x1p to x2p (chgp)
. gen chgp = x2p - x1p

. // scatter plot of change in percentile rank vs starting percentile
. scatter chgp x1p , xline(20 80)

. graph export scatter.pdf, as(pdf) replace
file scatter.pdf saved as PDF format

.
. // descriptives for change in percentile rank: all data, and then all
. // starting percentiles between 20 and 80
. summ chgp ,detail

                            chgp
-------------------------------------------------------------
      Percentiles      Smallest
 1%     .3354034       .0358658
 5%     1.407059       .0392838
10%     2.584104       .0401611       Obs               1,000
25%     5.424407       .0429602       Sum of wgt.       1,000

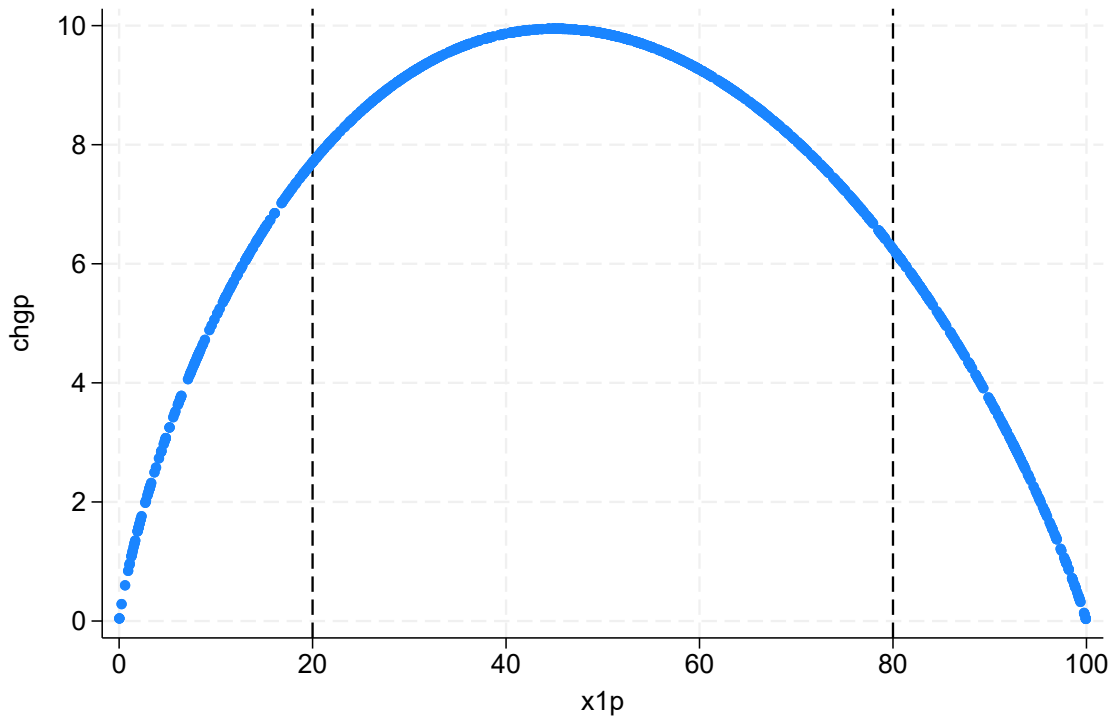
50%     7.995238                      Mean           7.053012
                        Largest       Std. dev.      2.778575
75%      9.41148       9.947632
90%     9.867844       9.947636       Variance       7.720477
95%     9.922853       9.947643       Skewness      -.8611846
99%     9.947395       9.947643       Kurtosis       2.585842
```

```
. summ chgp if x1p>20 & x1p<80, detail

                            chgp
-------------------------------------------------------------
      Percentiles      Smallest
 1%      6.43399       6.252701
 5%     6.810677       6.263039
10%     7.305954       6.297859       Obs                 611
25%     8.375114       6.341064       Sum of wgt.         611

50%     9.231464                      Mean           8.919597
                        Largest       Std. dev.      .9762341
75%     9.753323       9.947632
90%     9.914555       9.947636       Variance        .953033
95%     9.944225       9.947643       Skewness      -.9729678
99%     9.947624       9.947643       Kurtosis       3.012206

.
.
. // Close log and convert to PDF
.         capture log close
```


10
8
6
4
2
0
chgp
0
20
40
60
80
100
x1p